

RasC and Macropinocytosis (GO:0044351) in *Dictyostelium discoideum* — Focused Curation Report

Gene: rasC (DICDI) · **UniProt:** P32253 · **Hypothesis slug:** function-hypothesis-go-0044351 **Focus type:** function_assignment **Seed hypothesis:** *Dictyostelium discoideum* RasC participates in macropinocytosis (GO:0044351). Distinguish positive/negative regulatory contributions, direct machinery roles, compensatory Ras paralogs, and secondary growth effects. An increase in fluid uptake after deletion does not by itself establish lack of a role.

Executive Judgment

Verdict: Weakly supported / over-annotated for a direct role. There is **no primary experimental evidence that RasC directly participates in macropinocytosis**. Across the entire primary macropinocytosis literature in *Dictyostelium*, the Ras proteins repeatedly and specifically implicated are **RasG, RasS, and RasB** — never RasC. RasC's experimentally validated function is a distinct axis: RasGEFA(Aimless)→RasC→TORC2→PKB, driving cAMP relay (adenylyl cyclase activation) and chemotaxis. Direct paralog-substitution data show RasC **cannot** replace RasG in growth in suspension, a macropinocytosis-dependent process.

The current GO:0044351 annotation on P32253 carries the evidence code **IBA** (Inferred from Biological Ancestor, GO_Central) — a phylogenetic projection, not experimental support. Given that RasC is ~66% identical to RasG (the validated macropinocytosis regulator) and sits in the same catalytic Ras clade, the annotation is best explained as **clade-level paralog carry-over**.

Most important caveat (honoring the seed): Absence of a published rasC⁻ macropinocytosis assay means a *subtle or negative-regulatory* contribution cannot be formally excluded. The seed's warning — that increased fluid uptake after deletion would not by itself disprove a role — is methodologically valid, but here there is no such deletion-uptake dataset in evidence either way; the case against a *direct machinery* role rests on positive identification of other

paralogs plus a direct negative substitution result. This warrants **not** asserting a direct RasC macropinocytosis function while flagging the IBA term as a candidate for removal/generalization pending curator review.

Evidence Matrix

Citation (PMID)	Evidence type	Direction	Claim tested	Key finding	Context
30967009 (Williams, Paschke, Kay 2019)	Review of genetics + gain/loss-of-function	Competing / refutes direct RasC role	Which Ras drive macropinocytosis	"RasG and RasS are the key Ras proteins involved"; GefF implicated GEF	<i>D. discoideum</i> , growing cells
23843627 (Hoeller et al. 2013)	Direct assay: Ras-PI3K binding + single mutants	Competing / refutes	Which Ras act via PI3K in macropinocytosis	RasG and RasS bind PI3K1/2 & PI3K4; single rasG or rasS mutants have severe macropinocytosis defects	<i>D. discoideum</i> , vegetative
27821733 (Junemann et al. 2016)	Direct assay: effector regulation	Competing / refutes	Upstream Ras of macropinocytosis formin ForG	ForG directly regulated by RasB and RasG	<i>D. discoideum</i> , macroendocytosis
38263885 (Putar et al. 2024)	Mutant + localization	Competing / refutes	Major macropinocytosis Ras	"RasG is a major regulator of macropinocytosis"; IqgC loading depends on RasG	<i>D. discoideum</i>
20833893 (Bolourani et al. 2010)	Mutant paralog-substitution	Refutes (negative)	Can RasC substitute for RasG in growth in suspension (macropinocytic feeding)?	RasD fully substitutes; RasC is without effect	<i>D. discoideum</i> vegetative
17380187 (Kae et al. 2007)	Direct GEF specificity (in vitro + in vivo)	Qualifies (assigns RasC elsewhere)	RasC pathway specificity	RasGEFA activates RasC for adenylyl cyclase activation; RasG for chemotaxis	<i>D. discoideum</i> development
20493808 / 20660630 /	Direct assay / interaction	Qualifies	RasC effector	RasC is the upstream activator	<i>D. discoideum</i>

Citation (PMID)	Evidence type	Direction	Claim tested	Key finding	Context
27172998 (Charest, Cai, Khanna)				of TORC2→PKB for chemotaxis/cAMP relay	
25815683 (Bloomfield et al. 2015)	Forward genetics / mutant	Context	What limits fluid uptake	NF1 RasGAP loss increases fluid uptake via greater Ras/PI3K activity	<i>D. discoideum</i> axenic strains
UniProt P32253 (dictyBase/GO_Central)	Database annotation	Qualifies	Current annotation basis	GO:0044351 = IBA ; all experimental codes (IDA/IMP/IGI) are TORC2/cAMP/chemotaxis	Annotation record
24742374 (Chattwood et al. 2014)	Mutant / growth context	Qualifies (supports no essential RasC role)	Does feeding require RasC?	rasG ^{-C-} double-null cells are routinely grown in axenic medium (which depends on macropinocytic feeding), i.e. bulk fluid uptake persists without RasC	<i>D. discoideum</i>
This report (computed)	Computational (sequence)	Structural/ evolutionary	Paralog relatedness	RasC↔RasG 65.6%, ↔RasD 64.6%, ↔RasB 58.4%, ↔RasS 53.1% identity (global NW)	In-silico

GO Curation Implications

Lead (requires curator verification):

- **Term:** GO:0044351 macropinocytosis (BP), currently on P32253 via **IBA**.
- **Recommended action: Do not retain as an experimentally-styled assertion of RasC function.** Treat as **non-core / candidate for removal or generalization**. The experimental record supports RasG/RasS/RasB for this process, not RasC.
- Option A (preferred): **Remove** the IBA macropinocytosis annotation from RasC, or down-rank it, noting paralog carry-over from RasG.
- Option B (conservative): **Retain but flag** as phylogenetic-only (IBA) with an explicit note that no RasC-specific experimental evidence exists; do not promote to an experimental code.
- **Terms that ARE experimentally supported and should be emphasized as core:**
GO:1904841 TORC2 complex binding (IDA), GO:1904515 positive regulation of TORC2 signaling (IDA), GO:0010856 adenylate cyclase activator activity (IGI), GO:0030250 guanylate cyclase activator activity (IMP), GO:0043327 chemotaxis to cAMP (IMP), GO:0007188 adenylate cyclase-modulating GPCR signaling (IMP). These represent RasC's primary function.

"Protein binding" is not used as a recommendation; the informative core MF/BP terms above are supported.

GO Decision Table (lead — requires curator verification)

GO ID	Term (aspect)	Current code on P32253	Evidence for RasC specifically	Recommended curation action
GO:0044351	macropinocytosis (BP)	IBA (phylogenetic)	None experimental; paralog carry-over from RasG	Remove or flag as phylogenetic-only / non-core ; do not promote to experimental code
GO:1904841	TORC2 complex binding (MF)	IDA	Direct (RasC–TOR/TORC2 binding)	Retain — core
GO:1904515	positive regulation of TORC2 signaling (BP)	IDA	Direct	Retain — core
GO:0010856	adenylate cyclase activator activity (MF)	IGI	Genetic, RasC-specific	Retain — core
GO:0030250	guanylate cyclase activator activity (MF)	IMP	Mutant phenotype	Retain
GO:0043327	chemotaxis to cAMP (BP)	IMP	Mutant phenotype	Retain — core
GO:0007188	adenylate cyclase-modulating GPCR signaling (BP)	IMP	Mutant phenotype	Retain — core
GO:0031152	aggregation involved in sorocarp development (BP)	IMP	Mutant phenotype	Retain

Mechanistic Scope

- **Immediate molecular function of RasC:** a GTP-binding/GTPase small G-protein that, when activated by its cognate GEF **RasGEFA (Aimless)**, binds and activates **TORC2**, driving PKB/PKBR1 phosphorylation and adenylyl cyclase A activation (cAMP relay) and chemotaxis to cAMP.
 - **Macropinocytosis machinery (separate module):** organized around a plasma-membrane patch of PIP3 + active **RasG/RasS/RasB** and Rac, recruiting SCAR/WAVE→Arp2/3 and the formin ForG; regulated by GEFs (GefF, RasGEFX) and GAPs (NF1, RGBARG, IqgC, Leep2). RasC is not a documented member of this module.
 - **Downstream/secondary effects to keep separate from a direct role:** any effect of rasC deletion on growth or fluid uptake would be a *secondary* consequence (e.g., altered global Ras tone, compensatory RasD/RasG expression), not evidence of RasC acting in the macropinocytic cup.
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Conflicts and Alternatives

1. **Paralog confusion / clade projection (most likely):** RasC ~66% identical to RasG. IBA rules project ancestral function across the clade, sweeping RasC into "macropinocytosis" without species/paralog-specific data.
 2. **Shared reporter ambiguity:** Excitable-network and cup-imaging studies often use RBD (Ras-binding-domain) biosensors that detect active Ras collectively and cannot distinguish RasC from RasG/RasD; such images should not be read as RasC-specific localization to macropinocytic cups.
 3. **Compensation masking:** Because RasG (and RasD when de-repressed) can cover growth/cytokinesis, a rasC single deletion could appear neutral or even show increased uptake through compensatory paralog activity — the seed's caveat. This is an argument for caution, not for asserting a direct role.
 4. **Negative substitution result:** Bolourani 2010 is the cleanest conflict with the hypothesis — RasC explicitly fails to substitute for RasG in growth in suspension (macropinocytic feeding), whereas RasD succeeds.
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Knowledge Gaps

Gap	What was checked	Why it matters	Resolving evidence
No published rasC ⁻ (or rasC-GTP-locked) macropinocytosis / fluid-phase uptake assay	Multiple PubMed query variants (RasC + macropinocytosis/ pinocytosis/endocytosis/fluid-phase/growth/axenic) returned no direct RasC uptake assay; no study reports RasC-GTP binding the macropinocytosis effectors PI3K1/2/4 or ForG	Cannot formally exclude a subtle positive or negative-regulatory RasC contribution; also means the seed's "increased fluid uptake after deletion" scenario has no supporting dataset in evidence	TRITC-dextran fluid-uptake time course in rasC ⁻ , rasC ⁻ rasG ⁻ , and RasC(Q62L) vs WT
Whether RasC localizes to macropinocytic cups	UniProt CC terms list plasma membrane/leading edge/ vesicle but not cup-specific; imaging uses pan-Ras RBD probes	Localization to cups would be needed for a direct machinery role	RasC-specific (not RBD) live reporter co-imaged with PIP3/PHcrac and SCAR/ WAVE
Whether RasC-GTP binds the macropinocytosis PI3Ks (PI3K1/2, PI3K4)	Hoeller 2013 tested RasG/ RasS binding, not RasC	Direct effector binding would upgrade from IBA	In vitro Ras-PI3K RBD pulldowns including RasC
Basis of the IBA call	Confirmed evidence code = IBA on P32253	Determines whether curator should trust the annotation	Inspect the GO_Central PAINT tree / ancestral node for the projection

Discriminating Tests

1. **Quantitative fluid-phase uptake (TRITC-dextran/FITC-dextran) and macropinosome sizing** in WT vs rasC⁻ vs rasG⁻ vs rasC⁻rasG⁻, plus constitutively active RasC(Q62L). A direct positive role predicts reduced uptake in rasC⁻ and/or increased uptake with RasC(Q62L); a purely compensatory/secondary effect predicts little change (or increase) in rasC⁻ that is explained by paralog upregulation.
 2. **RasC-specific live-cell localization** (not RBD) relative to PIP3 and SCAR/WAVE cup markers.
 3. **Direct effector-binding assays** of RasC-GTP with PI3K1/2, PI3K4, and ForG, benchmarked against RasG/RasS/RasB.
 4. **Compensation controls:** measure RasG/RasD/RasS activation and protein levels in rasC⁻ to interpret any uptake phenotype.
 5. **Inspect the GO_Central PAINT annotation tree** to see whether the macropinocytosis node was experimentally anchored on RasG and projected to RasC.
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Curation Leads (require curator verification)

- **Candidate action change:** Flag/remove or mark-as-phylogenetic the RasC → GO:0044351 (macropinocytosis) IBA annotation; do not treat RasC macropinocytosis as core function.
- **Candidate references to cite in the review:**
 - PMID 30967009 — snippet to verify: *"genetic analysis suggests that RasG and RasS are the key Ras proteins involved."*
 - PMID 23843627 — *"RasG and RasS interact most strongly with PI3K1/2 and PI3K4, and single mutants of either Ras have severe macropinocytosis defects."*
 - PMID 27821733 — *"ForG is directly regulated in large-scale endocytosis by RasB and RasG."*
 - PMID 38263885 — *"RasG is a major regulator of macropinocytosis."*
 - PMID 20833893 — *"RasD is capable of totally substituting for RasG for cytokinesis and growth in suspension, whereas RasC is without effect."*
 - PMID 17380187 — *"RasG regulates chemotaxis and RasC is responsible for adenylyl cyclase activation."*

- **Suggested curator questions:** (a) Is the GO:0044351 IBA on RasC anchored on an experimentally-annotated RasG node? (b) Is there any unpublished dictyBase phenotype linking rasC^- to fluid uptake? (c) Should RasC's TORC2/cAMP annotations be emphasized as the core function set?
 - **Suggested experiments:** the discriminating tests above (fluid-uptake panel, RasC-specific imaging, RasC-PI3K binding).
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Provenance / Artifacts

- Computed Ras paralog global %identity matrix (Needleman-Wunsch): RasC↔RasG 65.6, ↔RasD 64.6, ↔RasB 58.4, ↔RasS 53.1; RasG↔RasD 82.5 (executed in this run).
- UniProt P32253 GO cross-reference dump confirming GO:0044351 = **IBA** and experimental codes limited to TORC2/cAMP/chemotaxis (executed in this run).
- Recorded knowledge-state findings #1–#3.

Computational results reported conservatively; simple global alignment used, no structural modeling performed. No fabricated assay results.
